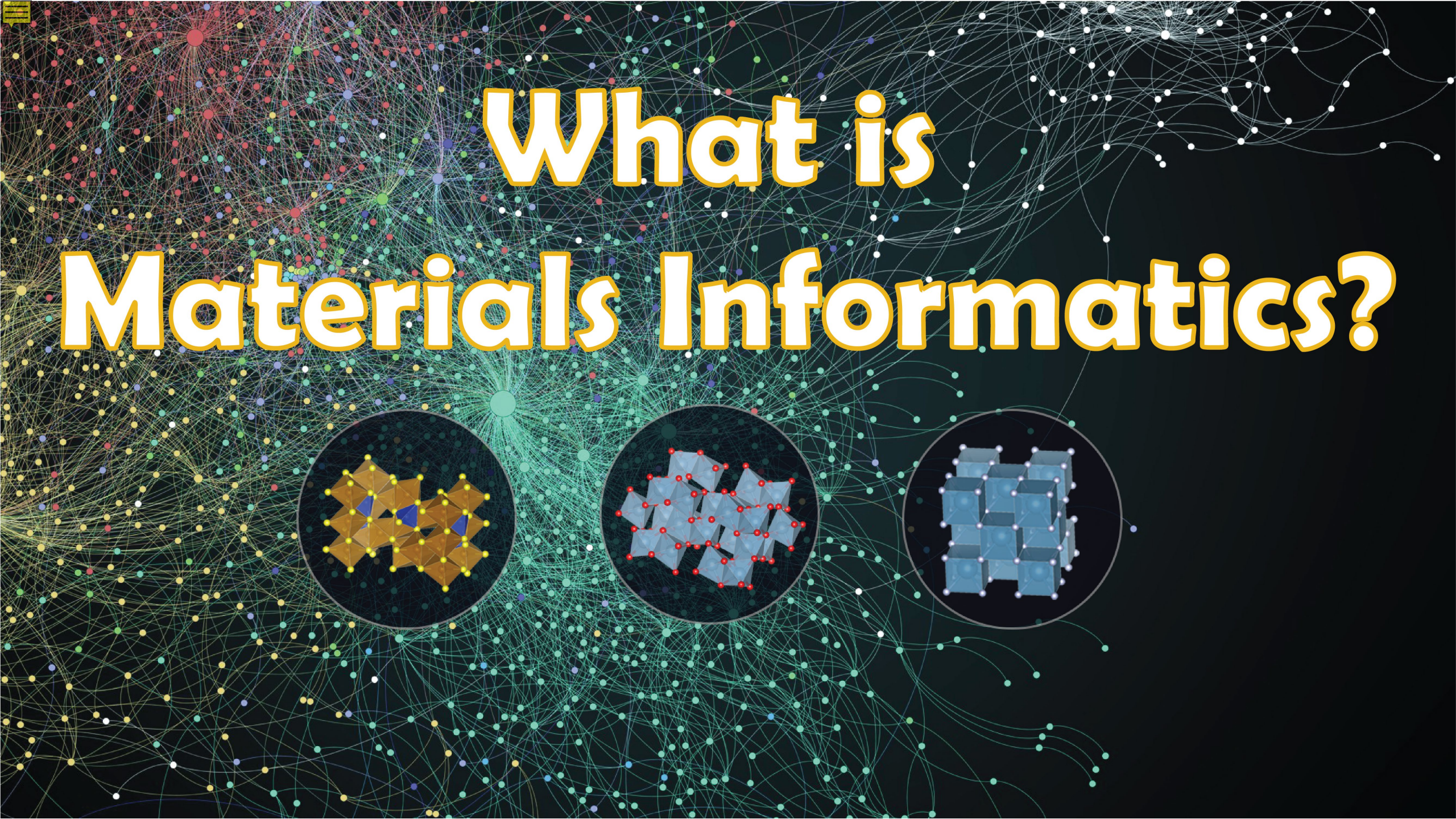


Materials Informatics

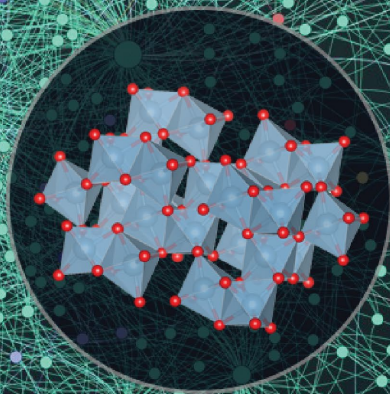
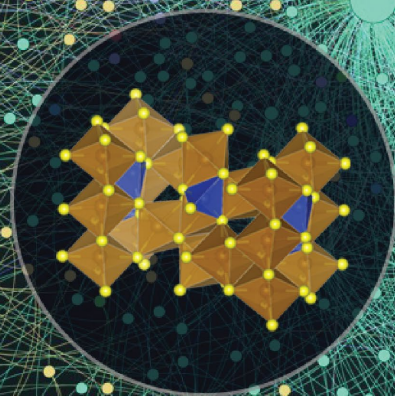


Taylor D. Sparks

University of Utah, Materials Science and Engineering Department

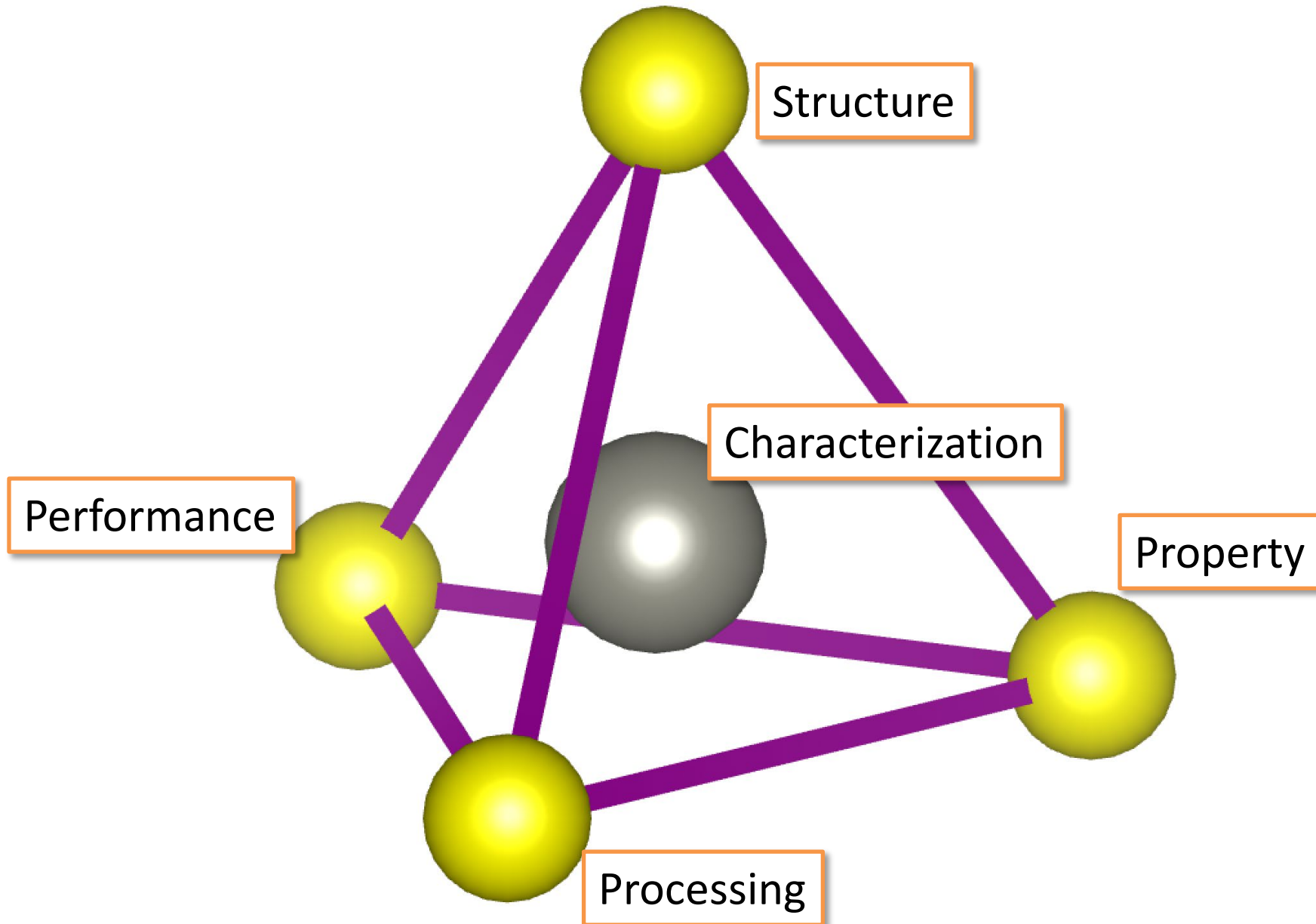


What is Materials Informatics?





Materials Informatics is data science applied to materials science



Materials Informatics is data science applied to materials science

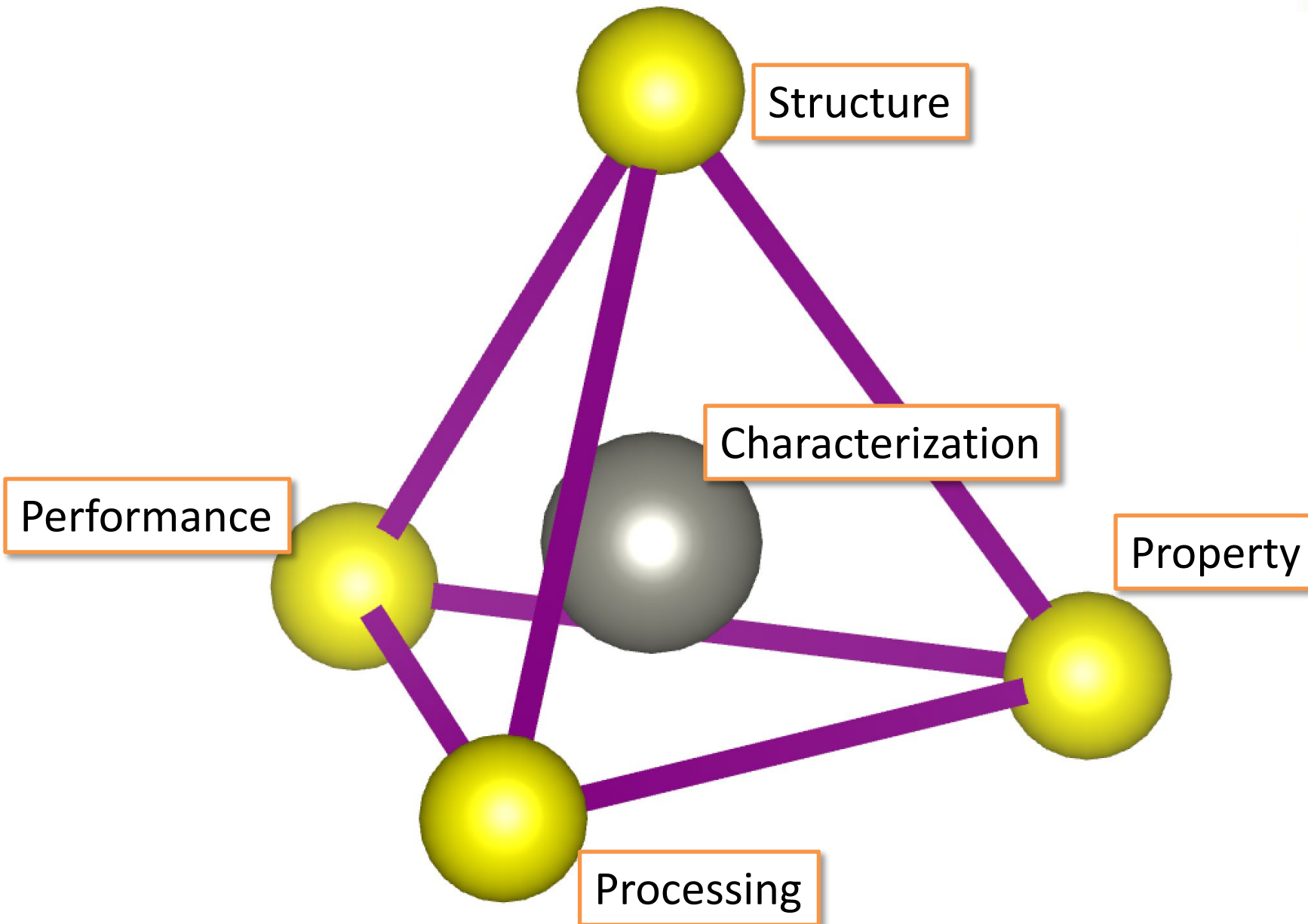
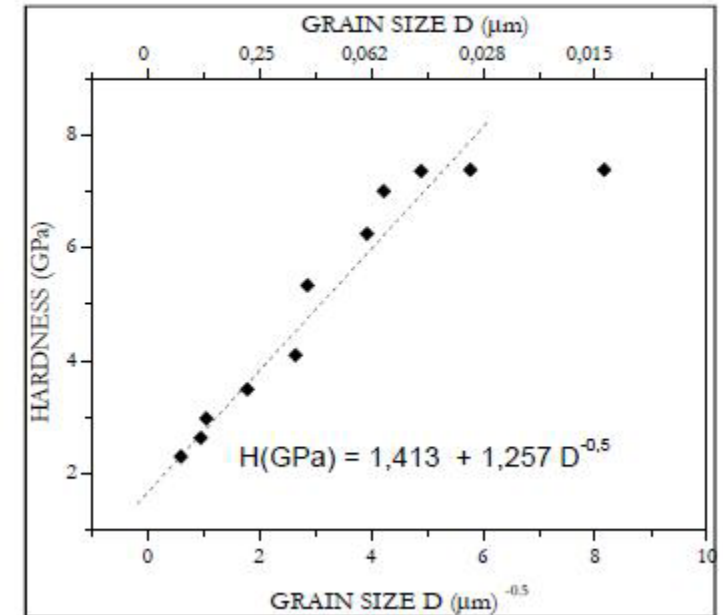


Figure 3. Hall-Petch ratio for 0.6%C steel



ML can extract highly complex trends in data!

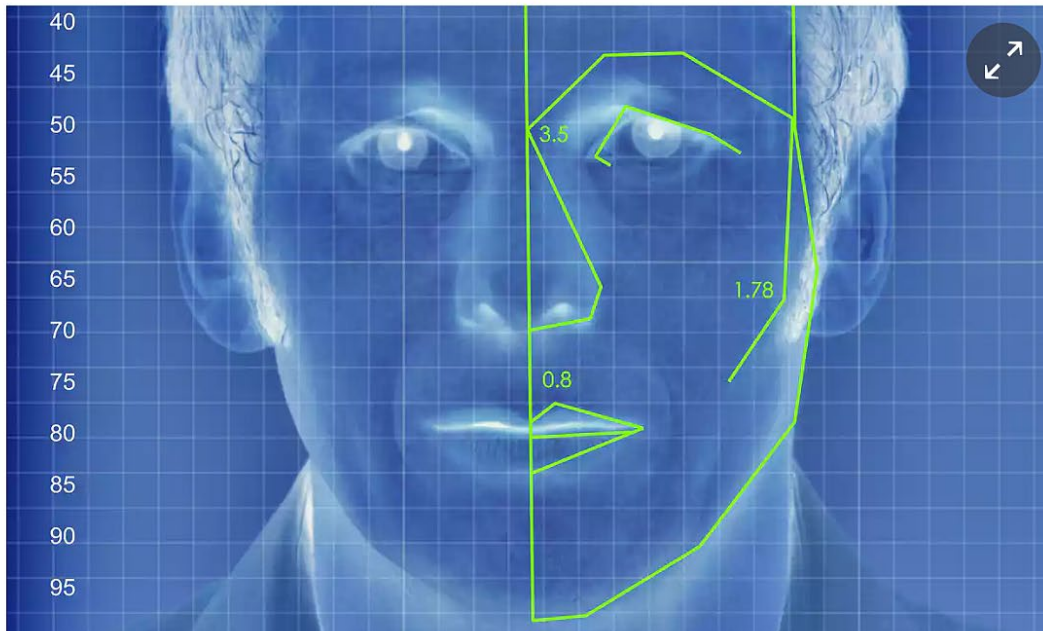
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New AI can guess whether you're gay or straight from a photograph

An algorithm deduced the sexuality of people on a dating site with up to 91% accuracy, raising tricky ethical questions



i An illustrated depiction of facial analysis technology similar to that used in the experiment. Illustration: Alamy

Most popular in US



Trump at UN: US may 'have no choice but to totally destroy North Korea' - live



Hurricane Maria: Storm grows in force to category 5 as Caribbean battered again - live



Where is Hurricane Maria heading? Mapping the path of destruction



Russian helicopter

ML has already made incredible advances in science

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**BLOG POST**
RESEARCH

30 NOV 2020

AlphaFold: a solution to a 50-year-old grand challenge in biology



SHARE



AUTHORS

 TAt The AlphaFold team

Proteins are essential to life, supporting practically all its functions. They are large complex molecules, made up of chains of amino acids, and what a protein does largely depends on its unique 3D structure. Figuring out what shapes proteins fold into is known as the “protein folding problem”, and has stood as a grand challenge in biology for the past 50 years. In a major scientific advance, the latest version of our AI system [AlphaFold](#) has been recognised as a solution to this grand challenge by the organisers of the biennial Critical Assessment of protein Structure Prediction ([CASP](#)). This breakthrough demonstrates the impact AI can have on scientific discovery and its potential to dramatically accelerate progress in some of the most fundamental fields that explain and shape our world.

A protein’s shape is closely linked with its function, and the ability to predict this structure unlocks a greater understanding of what it does and how it works. Many of the world’s greatest challenges, like developing treatments for diseases or finding enzymes that break down industrial waste, are fundamentally tied to proteins and the role they play.

BACK TO TOP ↑

ML has already made incredible advances in science

CASE
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CITRINE ACCELERATES DEVELOPMENT TIME FOR ADDITIVE MANUFACTURING

3D PRINTABLE AEROSPACE-GRADE ALLOY
DEVELOPMENT REDUCED FROM YEARS TO DAYS

CITRINE
INFORMATICS

HRL
LABORATORIES



Al 7A77
New alloy!

EXECUTIVE SUMMARY

FAST TIME TO MARKET

TWO YEARS
AFTER RESEARCH
PUBLICATION -
COMMERCIALIZED
WITH NASA AS THE
FIRST CUSTOMER

TIME SAVED

EXPERIMENTAL LAB
WORK REDUCED
FROM YEARS TO DAYS
DUE TO MATERIALS
INFORMATICS
APPROACH

PERFORMANCE IMPROVED

NEW ALLOY POWDER
RETAINS STRENGTH
WHEN USED IN
OFF-THE-SHELF 3D
PRINTING EQUIPMENT

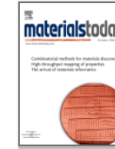
FIRST TO MARKET

FIRST ADDITIVE
ALLOY REGISTERED
BY THE ALUMINUM
ASSOCIATION

Materials Informatics is only a few decades old

materialstoday

Volume 8, Issue 10, October 2005, Pages 38-45



Review Feature

Materials informatics

Krishna Rajan 

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Seeking structure-property relationships is an accepted paradigm in materials science, yet these relationships are often not linear, and the challenge is to seek patterns among multiple lengthscales and timescales. There is rarely a single multiscale theory or experiment that can meaningfully and accurately capture such information. In this article, we outline a process termed ‘materials informatics’ that allows one to survey complex, multiscale information in a high-throughput, statistically robust, and yet physically meaningful manner. The application of such an approach is shown to have significant impact in materials design and discovery.

[<](#) **Previous** article in issue

Next article in issue [>](#)

In the early days... nobody knew what they were doing



Dan Ariely

January 6, 2013 · 🌐

Big data is like teenage sex: everyone talks about it, nobody really knows how to do it, everyone thinks everyone else is doing it, so everyone claims they are doing it...



13

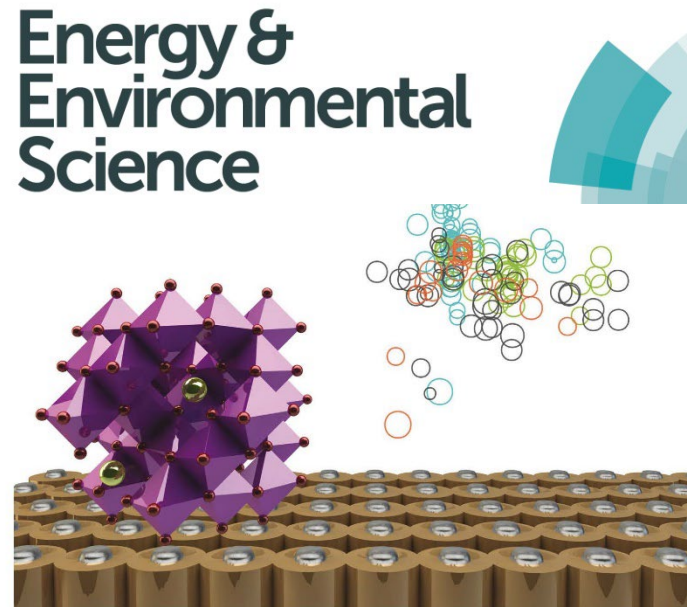


938



1.7K

We initially used “big data” to write analytical reviews

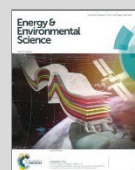


Showcasing research from the laboratories of Prof. Taylor Sparks at the University of Utah.

Performance and resource considerations of Li-ion battery electrode materials

Intercalation-based cathode materials follow a similar property-based tendency depending on their structure type. Simultaneous analysis of resource and performance parameters show that high energy materials are scarcer than high power materials. The website developed (<http://tomcat.eng.utah.edu/sparks/battery.jsp>) allows researchers worldwide to access our data and visualize any combination of parameters employing a unique visualization method.

As featured in:



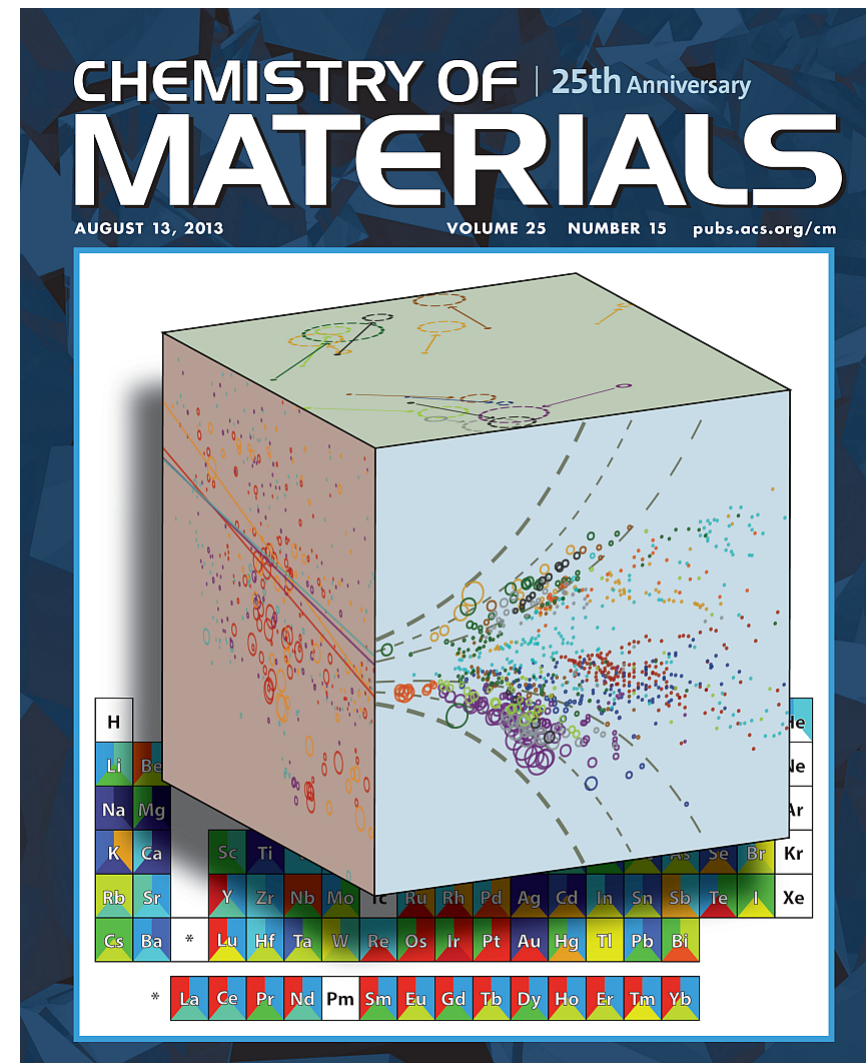
See Leila Ghadbeigi,
Taylor D. Sparks et al.,
Energy Environ. Sci., 2015, **8**, 1640.



www.rsc.org/ees

(Registered charity number: 201590)

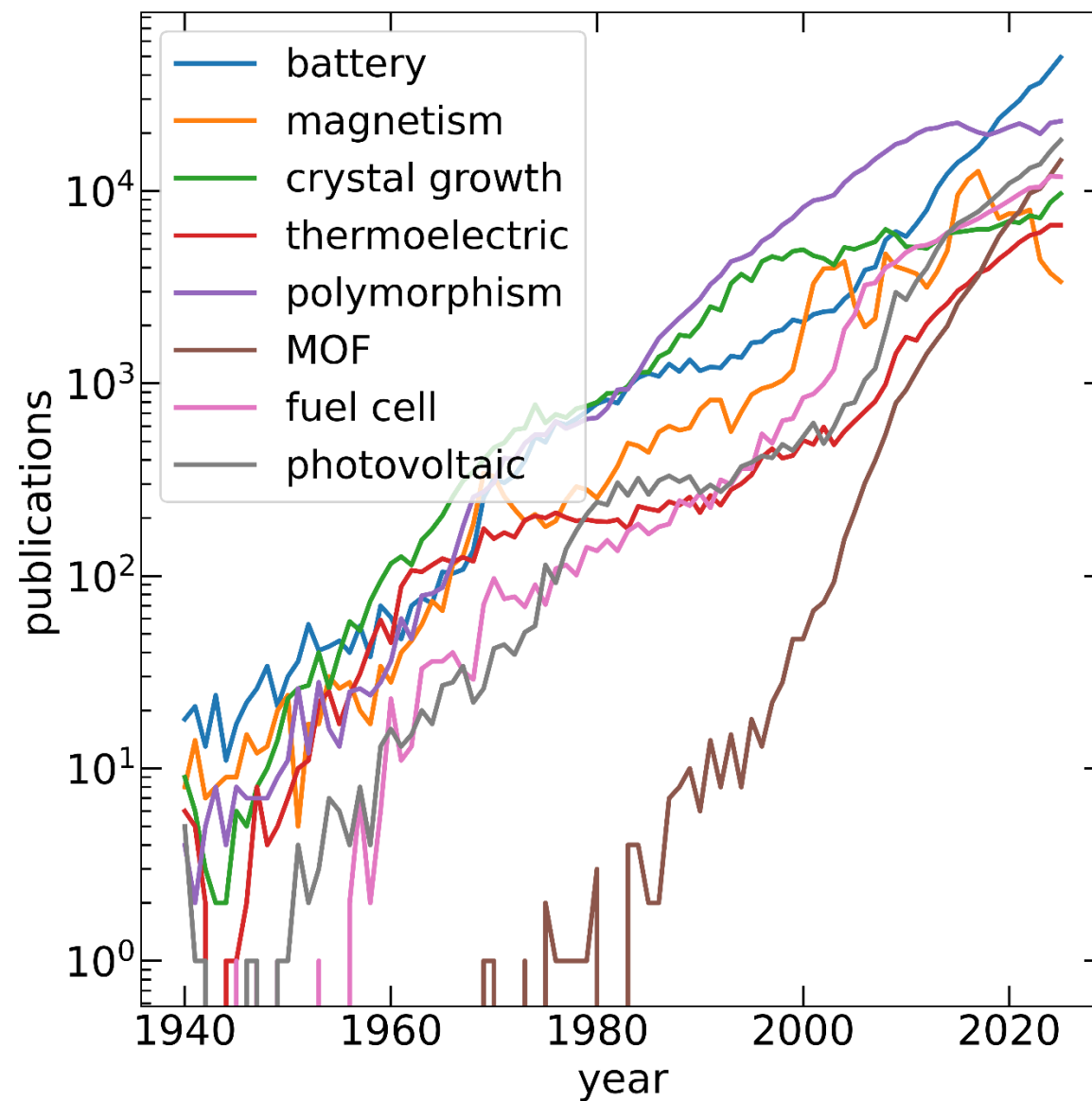
Sparks et al. (2016) *Script. Mat.*
Gaultois et al. (2013) *Chem Mat.*
Ghadbeigi et al. (2016) *Energy Environ. Sci.*
Gaultois et al. (2016) *APL Mat.*



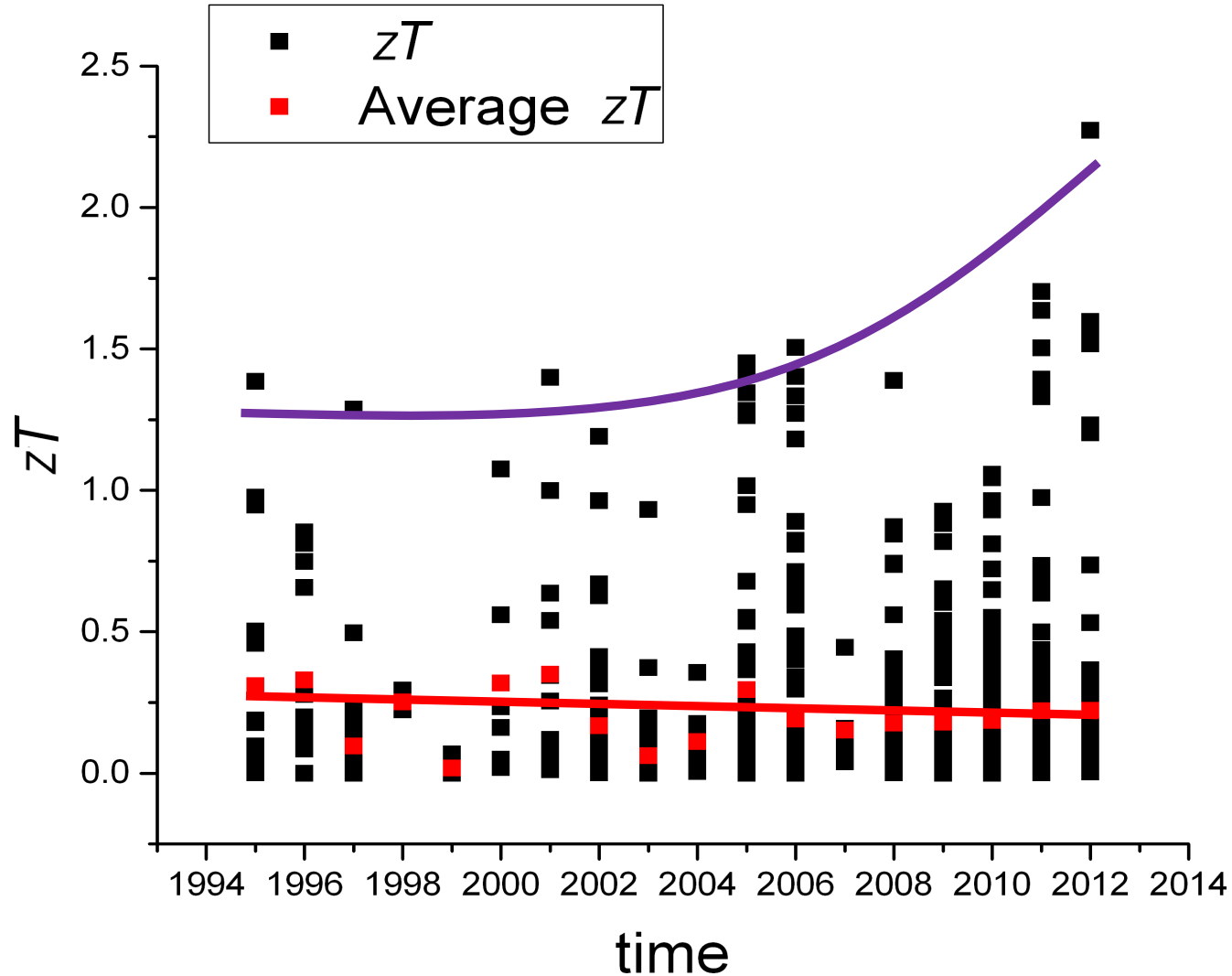
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Why do we need data-driven materials science?



Most materials improvements are incremental



Breakthroughs are the uncommon exception



Useful relationships can be learned from data directly!



Are there “materials genes” responsible for desired behavior?

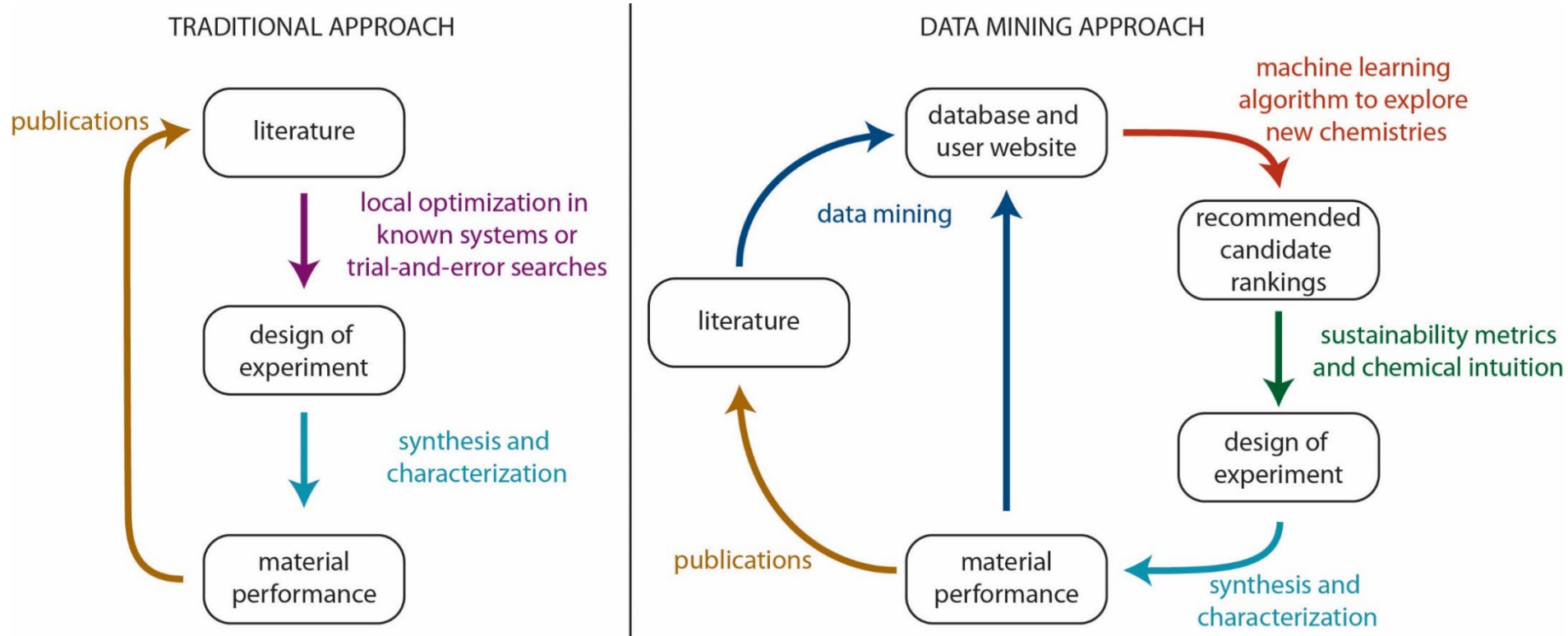
Materials Genome Initiative for Global Competitiveness

June 2011



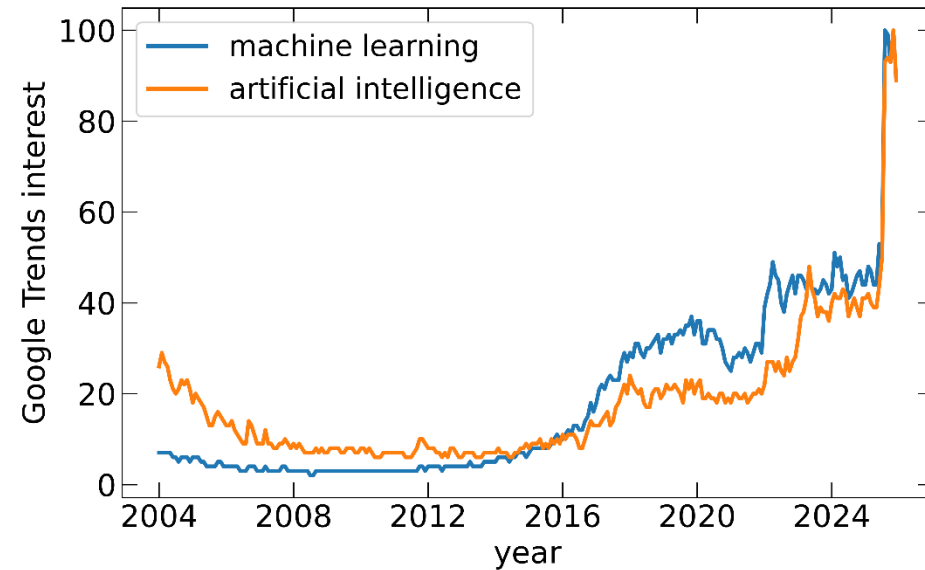
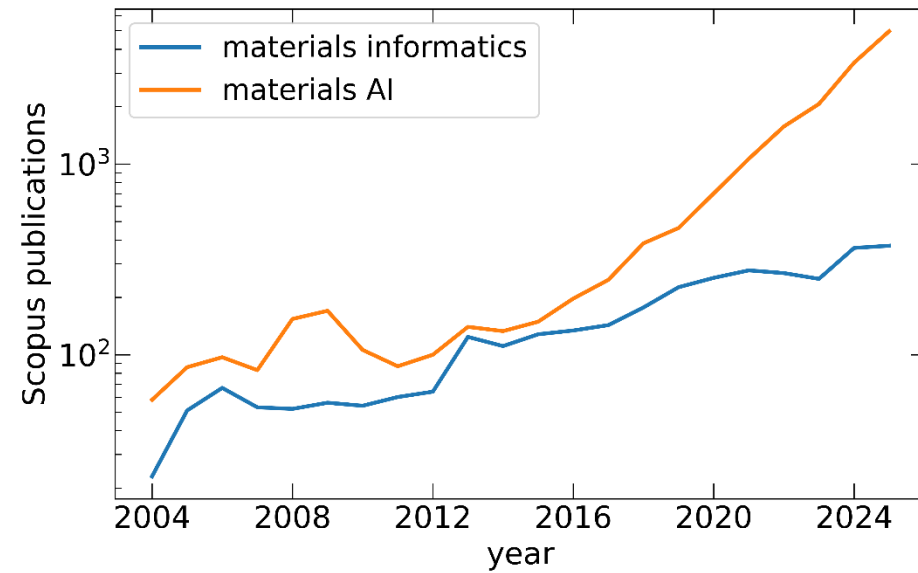
- Equip the next generation workforce
- Enable a paradigm shift in materials development
- Integrate experiments, computation, and theory
- Facilitate access to materials data

New tools of discovery are needed for “chemical whitespace”

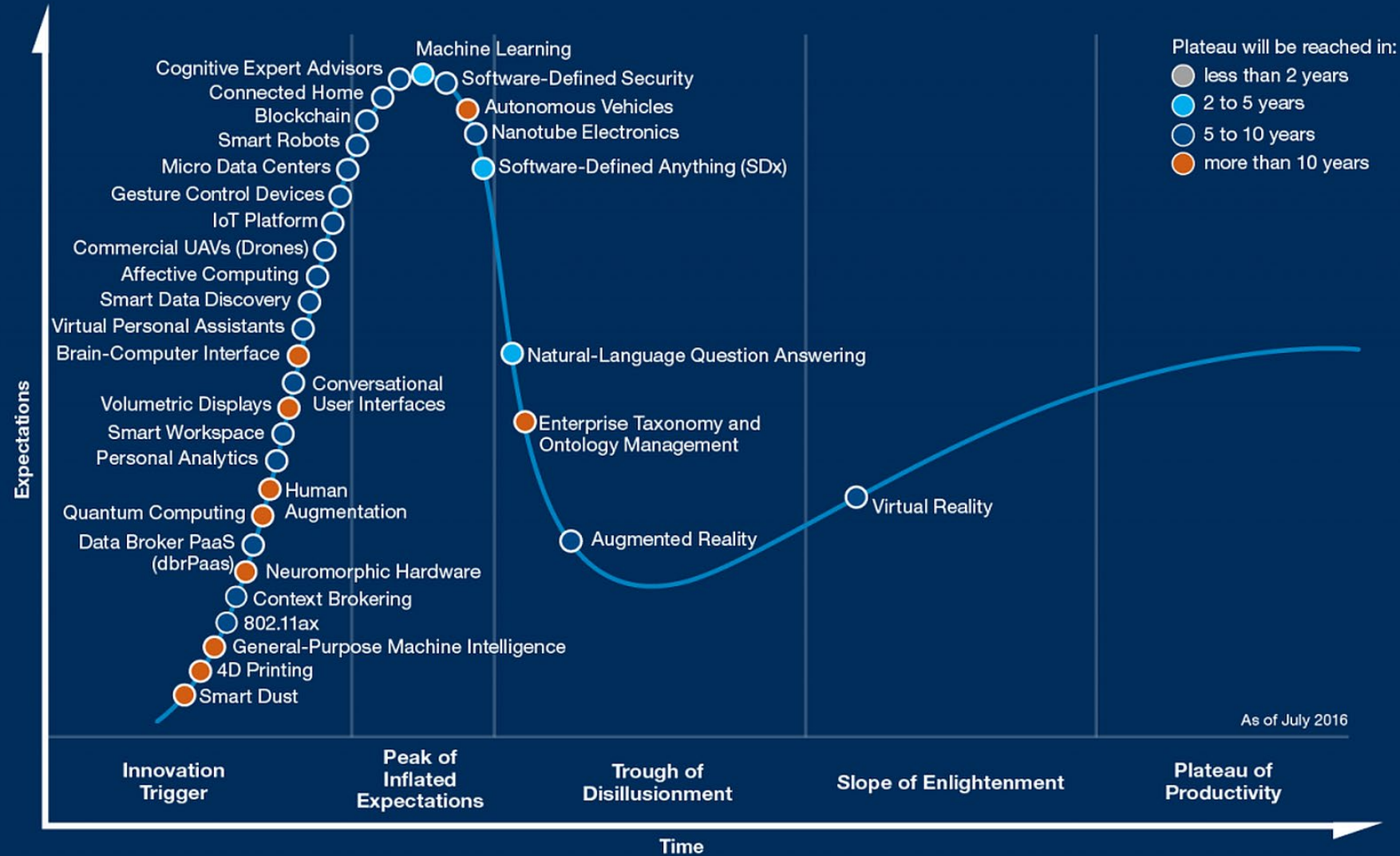




Is materials informatics a passing fad, or here to stay?

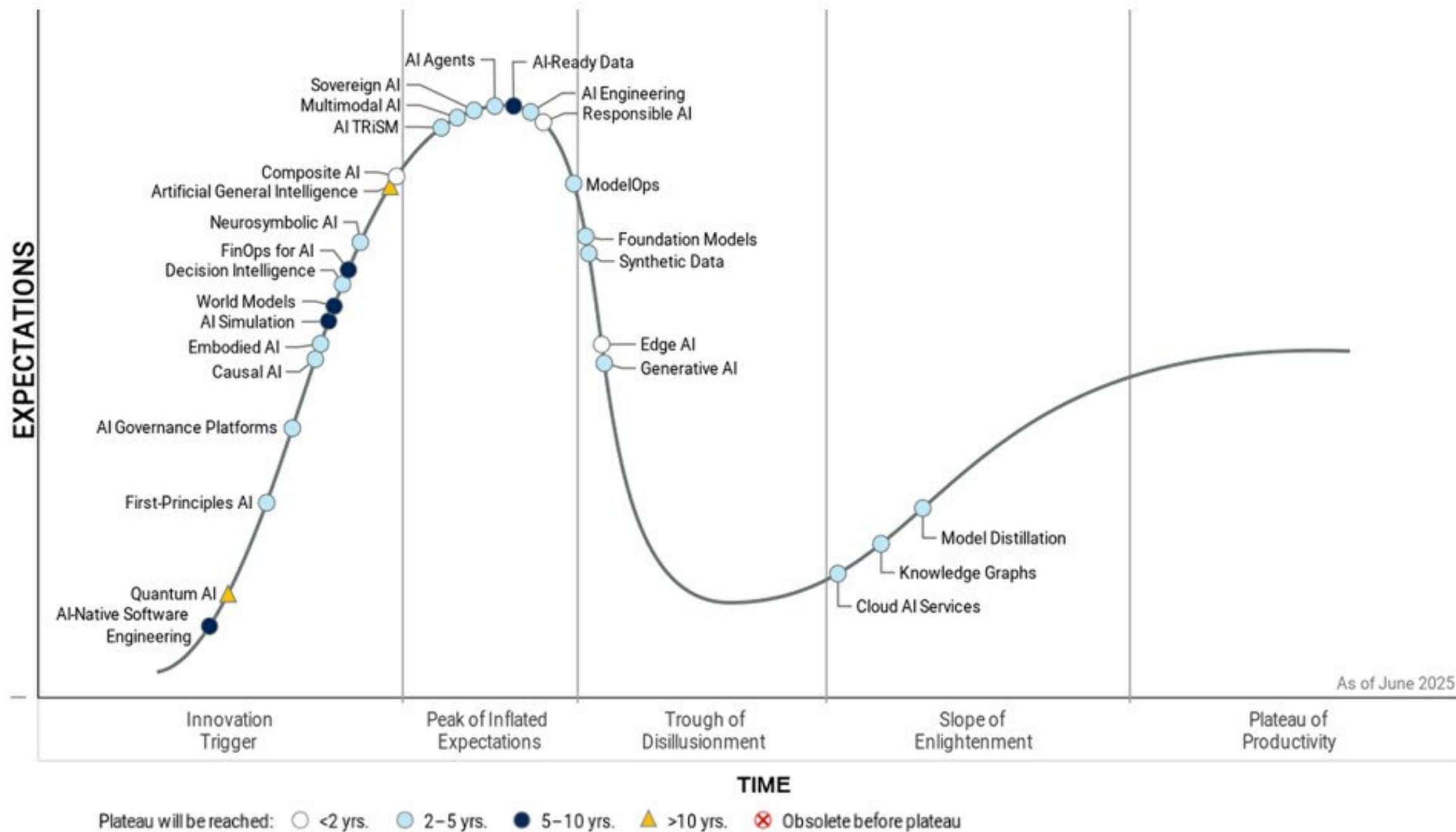


Gartner Hype Cycle for Emerging Technologies, 2016



Hype Cycle for Emerging Technologies, 2020





How are materials discovered?

